Statistics, Probability and Linear Programming
Solved Questions – Unit 1 & Unit 2

This PDF contains concise solved problems from Unit■1 and Unit■2 based on the uploaded question bank (pages 4–14).

Unit 1 – Least Squares Fit

Question: Fit a straight line $y = ax + b$ using least squares.

Solution: Normal equations:
$\Sigma y = na + b\Sigma x$
$\Sigma xy = a\Sigma x + b\Sigma x^2$
Solve simultaneously to obtain a and b.

Correlation Coefficient

Question: Define correlation coefficient.

Solution: Karl Pearson coefficient:
$r = [n\Sigma xy – \Sigma x\Sigma y] / \sqrt{[(n\Sigma x^2–(\Sigma x)^2)(n\Sigma y^2–(\Sigma y)^2)]}$

Regression Lines

Question: State regression equations.

Solution: Regression of y on x: $y–■ = b_{yx}(x–x■)$
Regression of x on y: $x–x■ = b_{xy}(y–■)$

Binomial Distribution

Question: Mean and variance of Binomial distribution.

Solution: For $X \sim B(n,p)$: Mean = np and Variance = npq where $q = 1–p$.

Poisson Distribution

Question: Mean and variance of Poisson distribution.

Solution: For $X \sim \text{Poisson}(\lambda)$: Mean = λ and Variance = λ.

Worked Binomial Example

Question: Probability of exactly 3 heads in 5 tosses.

Solution: Using $P(X=r)={}^{n}C_{r}p^{r}q^{n-r}$
$P(X=3)={}^{5}C_{3}(1/2)^{5}=10/32=0.3125$

Poisson Example

Question: Average defects = 2. Find probability of exactly 3 defects.

Solution: $P(X=3)=e^{-2}2^3/3! = 0.1804$

Normal Distribution

Question: State pdf of normal distribution.

Solution: $f(x)=1/(\sigma\sqrt{2\pi}) \cdot \exp[-(x-\mu)^2/(2\sigma^2)]$

Standard Normal Variable

Question: Define standard normal variable.

Solution: $Z=(X-\mu)/\sigma$ follows standard normal distribution with mean 0 and variance 1.

Exponential Distribution

Question: State pdf and mean.

Solution: $f(x)=\lambda e^{-\lambda x}$, $x>0$
Mean = $1/\lambda$

Uniform Distribution

Question: State pdf of Uniform(a,b).

Solution: $f(x)=1/(b-a)$, $a<x<b$

Gamma Distribution

Question: State mean and variance of Gamma distribution.

Solution: If $X\sim\text{Gamma}(\alpha,\beta)$: Mean = α/β and Variance = α/β^2

Joint Probability

Question: Define joint probability distribution.

Solution: A function giving probabilities of simultaneous occurrence of X and Y values.

Marginal Density

Question: Define marginal density function.

Solution: $f_X(x)=\int f(x,y)dy$ and $f_Y(y)=\int f(x,y)dx$

Conditional Probability Density

Question: Define conditional density.

Solution: $f(x|y)=f(x,y)/f_Y(y)$

Worked Normal Example

Question: If $X\sim N(50,10^2)$, find $P(X<60)$.

Solution: $Z=(60-50)/10=1$.
$P(Z<1)=0.8413$

Worked Exponential Example

Question: Mean lifetime = 5 years. Find $P(X>3)$.

Solution: $\lambda=1/5$.
$P(X>3)=e^{-3/5}=0.5488$

Worked Uniform Example

Question: $X\sim U(0,10)$. Find $P(2<X<6)$.

Solution: $P=(6-2)/(10-0)=4/10=0.4$

Joint Distribution Example

Question: If $f(x,y)=x+y$ on $0<x<1$, $0<y<1$, write marginal of X.

Solution: $f_X(x)=\int$■$^1(x+y)dy = x + 1/2$

Prepared from the uploaded MSRIT Statistics, Probability and Linear Programming question bank (Unit 1 & Unit 2).